Rop Aron K

Contact Information

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Education

Ph.D., Chemistry, University of Debrecen, Debrecen, Hungary — expected **Sep 2028**

Dissertation Advisor: Prof Dr. Kéki Sándor

Dissertation Title: *Intelligent Block Copolymers of Acrylamide Derivatives*

Msc., Chemistry, University of Debrecen, Debrecen, Hungary — **July 2023**

Dissertation Advisor: Assoc. Prof Nagy Tibor

Dissertation Title: *Self-Assembly of Pluronic-Type Amphiphilic Block Copolymers*

Bsc., Industrial Chemistry, Jomo Kenyatta University of Agriculture and Technology,Kenya **Nov 2018**

Degree conferred with First Class honors.

Employment

PhD Research Associate, University of Debrecen, Debrecen, Hungary — **Sep 2024–Present**

Research Interests

General relativity (GR), gravitation, and astrophysical phenomena which can elucidate gravity. One major theme is pushing numerical and analytical gravitational-wave (GW) predictions to the precision frontier in advance of next-generation observatories. A second major theme is using GWs to test GR against beyond-GR models, in both theory-independent and theory-dependent models. This involves numerical relativity and renormalization methods applied to specific effective field models for beyond-GR theories.

Honors and Awards

Stipendium Hungaricum , Tempus Public foundation, — **2021**

Teaching Experience

Professor, University of Mississippi

Phys. 213, General physics I — **Spring 2021**

Guest Lecturer, California Institute of Technology

Ph236, General relativity — **Fall 2017**

Ph237, Gravitational Waves — **Spring 2016**

Guest Lecturer, Massachusetts Institute of Technology

8.901, Graduate Astrophysics I — **Spring 2011**

Teaching Assistant, Massachusetts Institute of Technology

8.942, Cosmology — **Fall 2011**

8.901, Graduate Astrophysics I — **Spring 2011**

8.286, The Early Universe — **Fall 2009**

Teaching Assistant, California Institute of Technology

Ph 7, Nuclear and Quantum Physics Lab **Spring 2005**

Ph 5, Analog Electronics for Physicists **Fall 2004**

Mentoring/ Supervision

Postdoctoral researchers

Károly Csukás **Fall 2021–Summer 2024**

Now a postdoc at the Wigner Research Centre for Physics

José Tomás Gálvez Ghersi **Fall 2019–Spring 2021**

Now faculty at Universidad de Ingeniería y Tecnología, Peru

Graduate students

David Bronicki, University of Mississippi **Fall 2019–Summer 2023**

Subhayu Bagchi, University of Mississippi **Fall 2019–present**

Aniket Khairnar, University of Mississippi **Fall 2019–present**

Akshay Khadse, University of Mississippi **Fall 2018–Summer 2024**

Now a lecturer at Georgia College & State University

Lorena Magaña Zertuche, University of Mississippi **Fall 2018–Summer 2024**

Now a postdoc at NBI, Copenhagen, Denmark

Joe Rivest, University of Mississippi **Fall 2018–Summer 2024**

Sashwat Tanay, University of Mississippi **Fall 2018–Summer 2022**

Now a lecturer at UTEC, Lima, Peru

Maria (Masha) Okounkova, Caltech **Fall 2015–Summer 2019**

Now faculty at Pasadena City College

Baoyi Chen, Caltech **Fall 2016–Summer 2018**

Undergraduate students

Ayoade-Alabi Elizabeth Adunola, Chemical Engineering **2025**

Nassozi Nicole Andrea Crystal, Chemical Engineering **2025**

Publication Summary

Most important Publications —

1. Gergő Róth, Ákos Kuki, **Aron Kipyegon Rop**, Lajos Nagy, Zuura Kyzy Kaldybek, Máté Benedek, Miklós Zsuga, Tibor Nagy, Sándor Kéki(2025). *Rapid Copolymer Analysis of Unresolved Mass Spectra by Artificial Intelligence*,Analytical Chemistry 2025 97 (**36**), 19801-19808

Conference Proceedings

4. Róth, G. , Kuki, Á. , Nagy, T. , **Rop, AK** , Zsuga, M. , Nagy, L. , Kaldybek, KZ , Benedek, M. , Kéki, S. :*Application of algorithms and neural networks to evaluate block copolymer mass spectra* In: XXXI. International Conference on Chemistry = 31st International Conference on Chemistry, Transylvanian Hungarian Technical Scientific Society, Cluj-Napoca, 48, **2025**, (ISSN 2734-7109)

3. **Rop, AK** , Róth, G. , Pardi-Tóth, VC , Zsuga, M. , Nagy, T. , Kéki, S. : *Rapid Synthesis of Thermoresponsive PNIPA-b-PNAM Block Copolymers.* In: Young Researchers' International Conference on Chemistry and Chemical Engineering (YRICCCE V), Babeș-Bolyai University, Faculty of Chemistry and Chemical Engineering, Cluj., 50, **2025**.

2. **Rop, AK** , Nagy, T. , Róth, G. , Kuki, Á. , Ayoade-Alabi, EA , Nassozi, NAC , Zsuga, M. , Kéki, S. : *Design and Synthesis of Thermoresponsive Copolymers.* In: XXXI. Nemzetkogő Kegyészkonferencia = 31st International Conference on Chemistry, Transylvanian Hungarian Technical Scientific Society, Cluj-Napoca, 47, **2025**, (ISSN 2734-7109)

1. **Rop, AK** , Nagy, T. , Kuki, Á. , Kéki, S. : *Self-Assembly of Pluronic-Type Amphiphilic Block Copolymers.* In: 4th Young Researchers‘ International Conference on Chemistry and Chemical Engineering YRICCCE IV - **2023**: Program and Book of Abstracts, Hungarian Chemical Society, Debrecen., 68, 2023. ISBN: 9786156018168

Referees

Scott A. Hughes, Professor of Physics, Massachusetts Institute of Technology
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Urbana, IL 61801-3003
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Yanbei Chen, Professor of Physics, California Institute of Technology
TAPIR 350-17, Caltech
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